Solution 4

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Remark:If you find any mistakes, typos, or have any other feedback, please let me know.

Problem 1

Let $X = X_1 + X_2$, where X_1 and X_2 are independent r.v.s taking values in $\mathbb{R}^d$ with $X_1 \sim N(a, \Sigma)$ and $X_2 \sim Poisson(\lambda, \mu)$. Prove that the characteristic function of X is

$$\phi_X(u) = \exp\left(ia^T u - \frac{1}{2}u^T \Sigma u + \int_{\mathbb{R}^d} \lambda \left(e^i u^T y - 1\right) \mu(dy)\right),$$

Proof. Given $X = X_1 + X_2$ where X_1 and X_2 are independent.

For $X_1 \sim N(a, \Sigma)$, the characteristic function is:

$$\phi_{X_1}(u) = \exp\left(ia^T u - \frac{1}{2}u^T \Sigma u\right)$$

Let $X_2 = \sum_{j=1}^{N} Y_j$, where $N \sim \text{Poisson}(\lambda)$ and $Y_j \sim \mu$ are i.i.d. random variables. By applying the Law of Total Expectation (conditioning on N), we have:

$$\begin{aligned}
\phi_{X_2}(u) &= E\left[\exp\left(iu^T \sum_{j=1}^{N} Y_j\right)\right] \\
&= \sum_{n=0}^{\infty} P(N = n) E\left[\exp\left(iu^T \sum_{j=1}^{n} Y_j\right)\right] \\
&= \sum_{n=0}^{\infty} e^{-\lambda} \frac{\lambda^n}{n!} \left(\phi_Y(u)\right)^n \\
&= e^{-\lambda} \sum_{n=0}^{\infty} \frac{(\lambda \phi_Y(u))^n}{n!} \\
&= e^{-\lambda} e^{\lambda \phi_Y(u)} = \exp\left(\lambda(\phi_Y(u) - 1)\right).
\end{aligned} \tag{1}$$

By substituting the integral representation $\phi_Y(u) = \int_{\mathbb{R}^d} e^{iu^T y} \mu(dy)$ and using the fact that $\mu(\mathbb{R}^d) = 1$, we obtain:

$$\phi_{X_2}(u) = \exp\left(\int_{\mathbb{R}^d} \lambda(e^{iu^T y} - 1)\mu(dy)\right). \tag{2}$$

Since independence holds,

$$\phi_X(u) = \phi_{X_1}(u)\phi_{X_2}(u)$$

$$\phi_X(u) = \exp\left(ia^T u - \frac{1}{2}u^T \Sigma u + \int_{\mathbb{R}^d} \lambda(e^{iu^T y} - 1)\mu(dy)\right)$$

This completes the proof.

□

Problem 2

Prove that every Lévy measure is σ-finite.

Proof. To prove that a Lévy measure ν on $\mathbb{R}^d$ is σ-finite, we recall its defining property: $\nu(\{0\}) = 0$ and $\int_{\mathbb{R}^d}(1 \wedge |y|^2)\nu(dy) < \infty$.

Consider the sequence of sets $E_n = \{y \in \mathbb{R}^d : |y| > 1/n\}$ for $n \in \mathbb{N}$. On each set E_n, the function $1 \wedge |y|^2$ is bounded below by a constant. Specifically, $1 \wedge |y|^2 \geq \min(1, 1/n^2) = 1/n^2$ for sufficiently large n. Thus:

$$\nu(E_n) = \int_{E_n} 1\,\nu(dy) \leq n^2 \int_{E_n} (1 \wedge |y|^2)\,\nu(dy) \leq n^2 \int_{\mathbb{R}^d} (1 \wedge |y|^2)\,\nu(dy) < \infty$$

Since $\mathbb{R}^d \setminus \{0\} = \bigcup_{n=1}^{\infty} E_n$ and $\nu(\{0\}) = 0$, the entire space $\mathbb{R}^d$ is a countable union of sets with finite ν-measure. Therefore, ν is σ-finite. □

Problem 3

Let X be a Lévy process with the characteristic (a, A, ν). Prove that

1. the measure $\tilde{\nu}$ given by $\tilde{\nu}(A) = \nu(-A)$ for all $A \in \mathscr{B}(\mathbb{R}^d)$ is also a Lévy measure;
2. $-X$ is also a Lévy process with the characteristic $(-a, A, \tilde{\nu})$.

Proof. (1)A Borel measure ν is a Lévy measure if $\nu(\{0\}) = 0$ and $\int_{\mathbb{R}^d}(1 \wedge |y|^2)\nu(dy) < \infty$.

1. Since $\nu(\{0\}) = 0$, we have $\tilde{\nu}(\{0\}) = \nu(-\{0\}) = 0$.
2. Set $z = -y$:

$$\int_{\mathbb{R}^d}(1 \wedge |y|^2)\tilde{\nu}(dy) = \int_{\mathbb{R}^d}(1 \wedge |-z|^2)\nu(dz) = \int_{\mathbb{R}^d}(1 \wedge |z|^2)\nu(dz) < \infty$$

Thus, $\tilde{\nu}$ is a Lévy measure.

(2) $\psi_X(u) = ia^T u - \frac{1}{2}u^T Au + \int(e^{iu^T y} - 1 - iu^T y\mathbf{1}_{|y|<1})\nu(dy)$. The symbol of $-X$ is given by $\psi_{-X}(u) = \psi_X(-u)$:

$$\begin{aligned}\psi_{-X}(u) &= ia^T(-u) - \frac{1}{2}(-u)^T A(-u) + \int_{\mathbb{R}^d}(e^{i(-u)^T y} - 1 - i(-u)^T y\mathbf{1}_{|y|<1})\nu(dy) \\ &= i(-a)^T u - \frac{1}{2}u^T Au + \int_{\mathbb{R}^d}(e^{iu^T(-y)} - 1 - iu^T(-y)\mathbf{1}_{|-y|<1})\nu(dy)\end{aligned}$$

Let $z = -y$. Since $\tilde{\nu}(dz) = \nu(dy)$, the integral becomes:

$$\int_{\mathbb{R}^d} (e^{iu^T z} - 1 - iu^T z \mathbf{1}_{|z|<1}) \tilde{\nu}(dz)$$

This matches the form of a Lévy process with triplet $(-a, A, \tilde{\nu})$. □

Problem 4

For the standard d_2-dimensional Brownian motion $B = \{B_t : t \geq 0\}$, determine the Lévy symbol and the characteristic of $X = \{at + PB_t : t \geq 0\}$, where $a \in \mathbb{R}^{d_1}$ and P is a $d_1 \times d_2$ matrix.

Proof. Let $X_t = at + PB_t$, where B_t is a standard Brownian motion in $\mathbb{R}^{d_2}$. The characteristic function of X_t for $u \in \mathbb{R}^{d_1}$ is given by:

$$\begin{aligned} \phi_{X_t}(u) &= E\left[\exp(iu^T(at + PB_t))\right] \\ &= \exp(ita^T u) E\left[\exp(i(P^T u)^T B_t)\right]. \end{aligned}$$

Since $B_t \sim N(0, tI_{d_2})$, the term $(P^T u)^T B_t$ is a scalar normal random variable with mean 0 and variance $t|P^T u|^2 = tu^T PP^T u$. Using the characteristic function of the normal distribution:

$$E\left[\exp(i(P^T u)^T B_t)\right] = \exp\left(-\frac{1}{2} tu^T PP^T u\right).$$

Thus,

$$\phi_{X_t}(u) = \exp\left(t\left(ia^T u - \frac{1}{2} u^T PP^T u\right)\right).$$

By the definition $\phi_{X_t}(u) = e^{t\Psi(u)}$, we identify:

1. Lévy Symbol:
$$\Psi(u) = ia^T u - \frac{1}{2} u^T (PP^T) u$$
2. Comparing this with the standard Lévy-Khintchine form (b, A, ν), we have:
   - Drift vector $b = a$
   - Diffusion matrix $A = PP^T$
   - Lévy measure $\nu = 0$

   The characteristic is $(a, PP^T, 0)$.

□

Problem 5

Prove the following statements.

1. A Poisson process is always a subordinator.
2. A compound Poisson process is a subordinator if $Z_n \geq 0$ a.s. for $\forall n \in \mathbb{N}$.

Proof. A Lévy process is a subordinator if its paths are almost surely non-negative and non-decreasing.

(1) Let $N = \{N_t : t \geq 0\}$ be a Poisson process with intensity $\lambda > 0$.

1. By definition, a Poisson process is a Lévy process (it has stationary, independent increments and starts at 0).

2. Consider arbitrary time points $0 \leq s < t$. The increment $N_t - N_s$ follows a Poisson distribution with parameter $\lambda(t-s)$:
$$N_t - N_s \sim \text{Poisson}(\lambda(t-s)).$$
The support of a Poisson distribution is the set of non-negative integers $\{0, 1, 2, \dots\}$. Therefore:
$$P(N_t - N_s \in \{0, 1, 2, \dots\}) = 1.$$
This implies:
$$N_t - N_s \geq 0 \quad \text{a.s.} \implies N_t \geq N_s \quad \text{a.s.}$$

(2) A compound Poisson process is defined as $X_t = \sum_{n=1}^{N_t} Z_n$.

1. Given $Z_n \geq 0$ a.s. for all $n \in \mathbb{N}$, and $N_t \geq 0$, the sum X_t is a sum of non-negative terms, so $X_t \geq 0$ a.s.

2. For $0 \leq s < t$, we have $X_t - X_s = \sum_{n=N_s+1}^{N_t} Z_n$. Since $N_t \geq N_s$ and $Z_n \geq 0$, the sum of these jump sizes is ≥ 0. Thus $X_t \geq X_s$ a.s.

Since the paths are non-negative and non-decreasing, X is a subordinator □

Problem 6

If X is a compound Poisson process, prove that for any $t > 0$,
$$\sum_{0 \leq s \leq t} |\Delta X_s| < \infty, \quad a.s.$$

Proof. Let X_t be a compound Poisson process defined as:
$$X_t = \sum_{k=1}^{N_t} Z_k,$$
where $N = \{N_t\}_{t \geq 0}$ is a Poisson process with intensity λ, and $\{Z_k\}_{k \geq 1}$ is a sequence of i.i.d. random variables representing the jump sizes.

The jumps of the process, $\Delta X_s = X_s - X_{s-}$, are non-zero only at the arrival times T_k of the Poisson process N. At these times, $\Delta X_{T_k} = Z_k$. For any other time s, $\Delta X_s = 0$.

Therefore,
$$\sum_{0 < s \leq t} |\Delta X_s| = \sum_{k=1}^{N_t} |Z_k|.$$

Since N_t follows a Poisson distribution with parameter λt, $P(N_t < \infty) = 1$. Since N_t is a finite integer a.s., and each $|Z_k|$ is finite a.s., the sum consists of a finite number of finite terms.

Consequently, the sum is finite almost surely:

$$\sum_{0<s\leq t} |\Delta X_s| < \infty \quad \text{a.s.}$$

$\square$